# ABHIJEETH MODI

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## Professional Summary

Senior Software Engineer with 4+ years of experience building and scaling distributed backend systems, event-driven architectures, and cloud-native platforms for enterprise contact center and campaign products. Expertise in Node.js, MongoDB, Redis, PostgreSQL, RabbitMQ, AWS, and real-time communication systems, with a strong track record of delivering high-availability production services, low-latency voice workflows, and observability-driven reliability improvements. Hands-on experience integrating Microsoft, OpenAI, Google, and Ultravox models for LLM-powered automation, streaming voice responses, and RAG.

## Work Experience

**Kore.ai Software India Pvt Ltd** **Aug 2022 – Present**
*Senior Software Engineer* *Hyderabad*

- Architected and scaled distributed backend systems for campaign and contact center platforms to sustain 99.99% uptime and 5M+ requests per second, using Node.js microservices, Redis-backed performance optimizations, RabbitMQ event orchestration, and 100+ concurrent Agenda jobs.
- Engineered the campaign execution platform to support 1M+ concurrent voice calls and millions of customer-agent interactions, enabling LLM-powered outreach flows that drove up to 55% business improvement across multiple enterprise customers.
- Built real-time voice streaming pipelines integrating Jambonz voice applications, SSE, sockets, streaming LLM responses, and TTS providers to deliver sub-500ms voice latency in production conversational workflows.
- Designed high-throughput Node.js and Express REST services with Socket.IO-based real-time delivery, handling 10K concurrent connections at p95 200ms with multi-tenant isolation and token-based authentication.
- Reduced end-to-end latency across campaign and contact center modules by 50% through asynchronous processing, Redis caching, database indexing, and request-path optimizations for high-volume production traffic.
- Integrated OpenSearch to power real-time analytics dashboards and cross-component data retrieval, designing indexing strategies across multiple identifiers that reduced search response times by up to 80%.
- Implemented centralized observability with Grafana and Prometheus, reducing incident rates across campaigns and contact center modules from 40% to 10% and lowering mean time to detect issues by 35%.
- Integrated Microsoft, OpenAI, Google, and Ultravox providers to ship LLM features including summarization, translation, sentiment analysis, recommendations, streaming voice responses, and production voice metrics such as TTFB and latency.
- Optimized AWS resource consumption through rightsizing of underutilized instances, improving infrastructure efficiency while preserving reliability and performance under peak production load.
- Designed and implemented smart, scheduled daily agenda notifications, periodic data ingestion using cron jobs and background schedulers, driving a 2× increase in user engagement and improving recurring application usage

## Skills

**Languages:** JavaScript, TypeScript, Python, SQL, C++
**Backend:** Node.js, Express.js, REST APIs, gRPC, WebSockets, Socket.IO, SSE, Microservices, Multi-Tenant Platform
**Databases and Messaging:** MongoDB, Redis, PostgreSQL, OpenSearch, Trino/Presto, RabbitMQ, Kafka
**Cloud and DevOps:** AWS, Docker, Nginx, Jenkins, Harness, CI/CD, Performance Optimization, Reliability Engineering
**Observability and Testing:** Grafana, Prometheus, OpenTelemetry, Jest, E2E Testing
**AI / ML :** LLM Integration, RAG, LangChain, LangGraph, OpenAI API, Microsoft Azure AI, Google AI, Ultravox, S2S, TTS
**Core Concepts:** System Design, Distributed Systems, Scalability, Caching, Elastic Search, Event-Driven Architecture, API Security

## Projects

**Enterprise GenAI RAG Chatbot** | *Node.js, LangChain, OpenAI, Pinecone* **2025**

- Designed and deployed an enterprise RAG pipeline handling multiple policy PDFs, optimizing chunking strategies and vector similarity search to achieve 94% retrieval precision and reduce hallucination by 40% through a custom prompt orchestration layer.
- Engineered a scalable modular API architecture with conversational memory that reduced context-retrieval latency by 35% to under 200ms and sustained throughput of 50+ concurrent user queries while maintaining grounded response generation.

## Education

**Indian Institute of Engineering Science and Technology (IIEST), Shibpur** **2018 – 2022**
*B.Tech in Electronics and Telecommunication Engineering* *CGPA: 8.67/10*

## Awards & Recognition

- **Outstanding Performer Q3 2025, Kore.ai** – Recognized for resolving complex production issues in voice campaigns and stabilizing high-scale customer deployments.
- **Global SpotLight Award Q4 2024, Kore.ai** – Awarded for key contributions to complex campaign dialer development and successful cross-team delivery of critical platform capabilities.
- **Top Performer Q4 2023, Kore.ai** – Recognized for improving backend reliability and observability across the voice campaigns module.